

Molecular-Dynamics Machines

Aiichiro Nakano

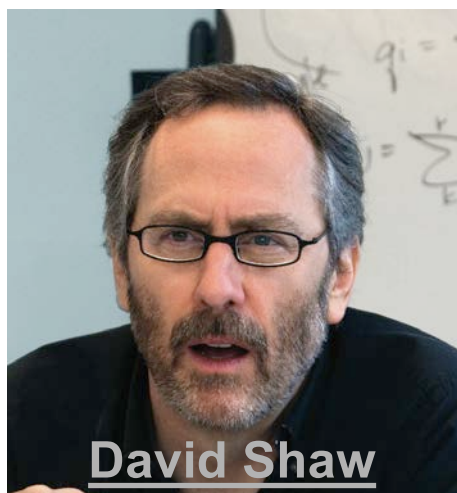
*Department of Computer Science
Department of Physics & Astronomy
Department of Quantitative & Computational Biology
University of Southern California*

Email: anakano@usc.edu

Why parallel MD? It's hot in computer science!



Anton: Computational Microscope



“... make all these discoveries because they were looking at the world in a different way.”

Named after Anton van Leeuwenhoek, who is often referred to as “the father of microscopy”

“... there’s still a lot of juicy, low-hanging fruit in this (molecular simulation) area ...”

A conversation with David E. Shaw, CACM 52(10), 49 ('09)

16 μ s/day simulation on 512 nodes
(5 μ s/step execution time)



D E Shaw Research

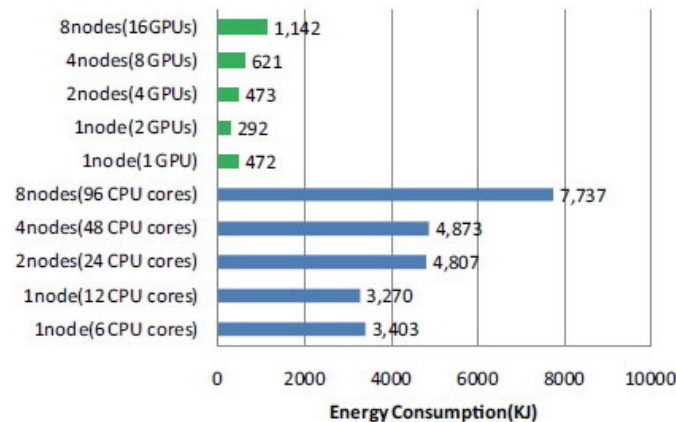
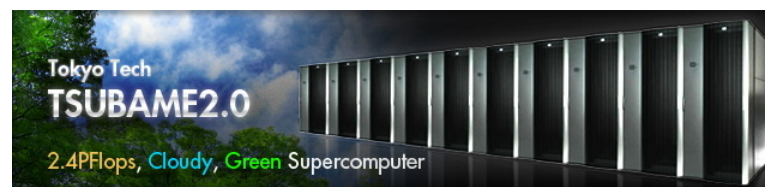
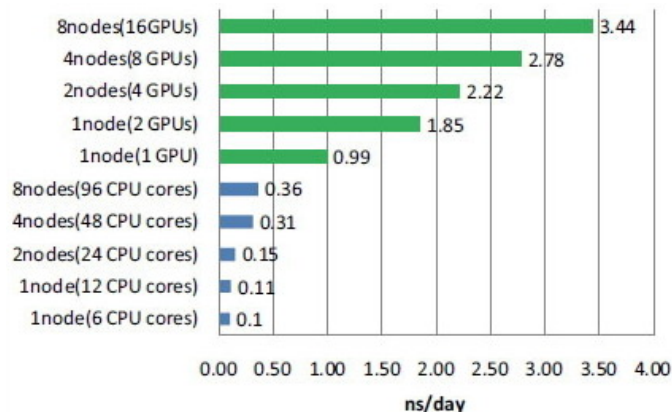
MD on GPU Clusters

GPU acceleration and other computer performance increases will offer critical benefits to biomedical science.

BY JAMES C. PHILLIPS AND JOHN E. STONE

Probing Biomolecular Machines with Graphics Processors

[CACM 52\(10\), 34 \('09\)](#)

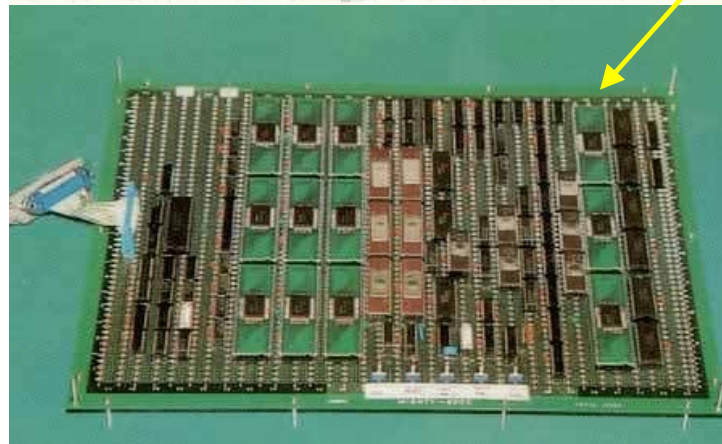
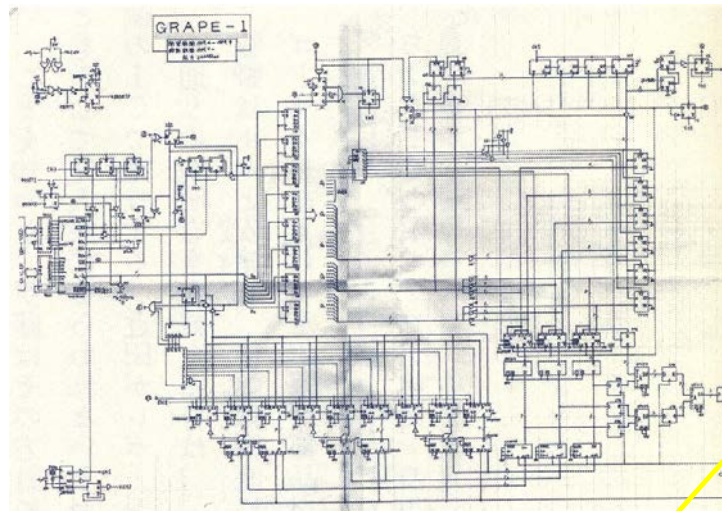


GRAPE 1 (\$2K, 1989)



Tomoyoshi Ito & comics he authored

- **GRAPE (GRAvity PipE)** = special-purpose computer for the gravitational N -body problem built by astrophysicists at Univ. of Tokyo
- **GRAPE 1** designed by a 1st-year Ph.D. student (with \$140K/year income)



Gravitational Pipeline

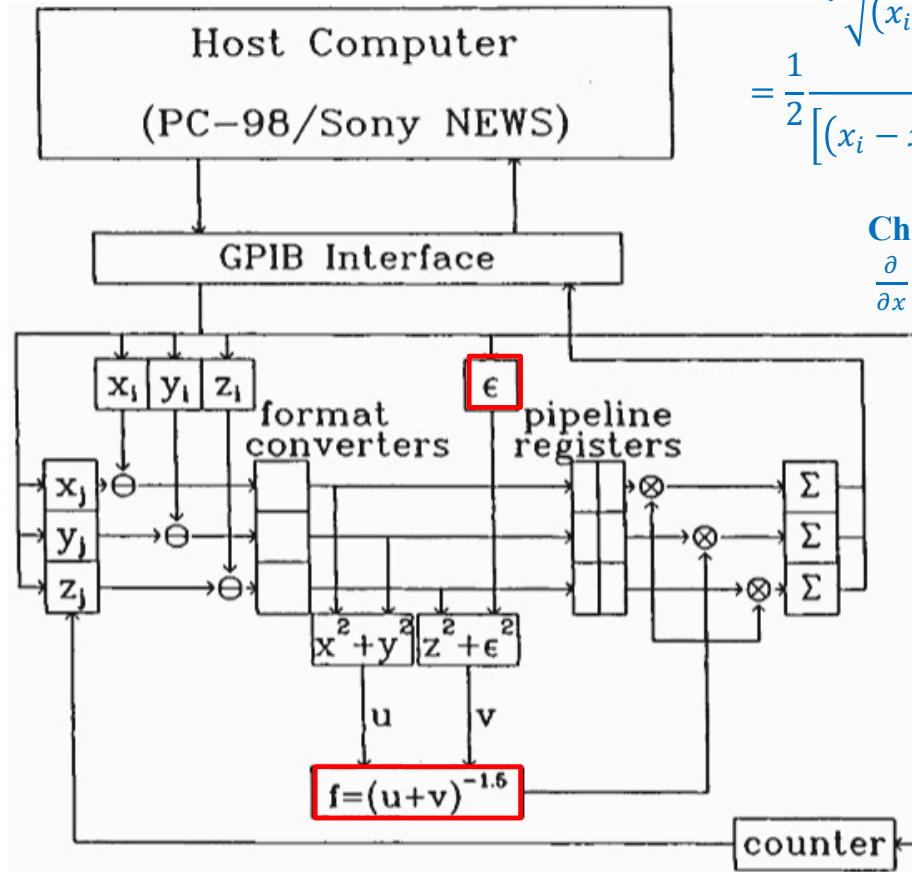
$$\frac{d^2 \mathbf{x}_i}{dt^2} = \mathbf{f}_i = \sum_j \frac{m_j (\mathbf{x}_j - \mathbf{x}_i)}{(r_{ij}^2 + \epsilon^2)^{3/2}}$$

$$-\frac{\partial}{\partial x_i} \frac{1}{\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2 + \epsilon^2}}$$

$$= \frac{1}{2} \frac{2(x_i - x_j)}{[(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2 + \epsilon^2]^{3/2}}$$

Chain rule:

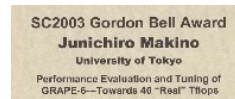
$$\frac{\partial}{\partial x} [f(x)]^{-1/2} = -\frac{1}{2} [f(x)]^{-3/2} \frac{\partial f}{\partial x}$$



- Small ϵ to avoid if $(i \neq j)$ & pipeline stall
- $x^{-3/2}$ was implemented as table look-up using read-only memory (ROM)

D. Sugimoto *et al.*,
Nature **345**, 33 ('90)

GRAPE & Gordon Bell Prizes



2003 Gordon Bell Prize, Special Achievement
Performance Evaluation and Tuning of GRAPE-6—Towards 40 'Real' Tflop/s
 Junichiro Makino, Hiroshi Daisaka, Eiichi Kokubo, Toshiyuki Fukushima



2001 Gordon Bell Prize, Winner, Peak Performance
A 11.55 Tflops simulation of black holes in a galactic center on GRAPE-6
 Junichiro Makino, Toshiyuki Fukushima



2000 Gordon Bell Prize, Winner, Peak Performance Category
A 1.34 Tflops simulation of black holes in a galactic center on GRAPE-6
 Junichiro Makino, Toshiyuki Fukushima, Masaki Koga

2000 Gordon Bell Prize, Winner, Peak Performance Category (tie with above)
1.34 Tflops Molecular Dynamic simulation for NaCl with a Special Purpose Computer: MDM (MD-GRAPE system)

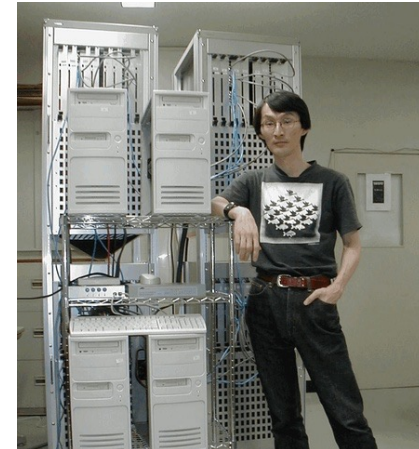
Tetsu Narumi, Ryutaro Susukita, Takahiro Koishi, Kenji Yasuoka, Hideaki Furusawa, Atsushi Kawai, Toshikazu Ebisuzaki

Green500 Rank	MFLOPS/W	Site*	Computer*	Total Power (kW)
1	1684.20	IBM Thomas J. Watson Research Center	NNSA/SC Blue Gene/Q Prototype	38.80
2+	1448.03	National Astronomical Observatory of Japan	GRAPE-DR accelerator Cluster, Infiniband	24.59
2	958.35	GSIC Center, Tokyo Institute of Technology	HP ProLiant SL390s G7 Xeon 6C X5670, Nvidia GPU, Linux/Windows	1243.80
3	933.06	NCSA	Hybrid Cluster Core i3 2.93Ghz Dual Core, NVIDIA C2050, Infiniband	36.00
4	828.67	RIKEN Advanced Institute for Computational Science	K computer, SPARC64 VIIIfx 2.0GHz, Tofu interconnect	57.96

www.green500.org (Nov. '10)

Gordon Bell Prize, commonly referred to as the Nobel Prize of Supercomputing [Wikipedia].

J. Makino & Grape 6 (2001)



1999 Gordon Bell Prize, Price Performance, First Prize
 Astrophysical N-body simulation
 144 Glops / \$ 1 M on custom-built GRAPE-5 32-processor system
 Atsushi Kawai, Toshiyuki Fukushima, and Junichiro Makino



1996 Gordon Bell Prize, Performance, Honorable Mention
 Simulation of the motion of 780,000 stars
 333 Gflops using the Grape-4 machine w/ 1,269 processors
 Junichiro Makino, Toshiyuki Fukushima



1995 Gordon Bell Prize, First Place, Special Purpose Machines
 Simulation of the Motion of 10,000 Stars
 112 Gflops using the Grape-4 machine with 288 processors
Astrophysical N-body Simulations on GRAPE-4 Special-Purpose Computer
 Junichiro Makino, Makoto Tajiri

Enabling Science by Hardware

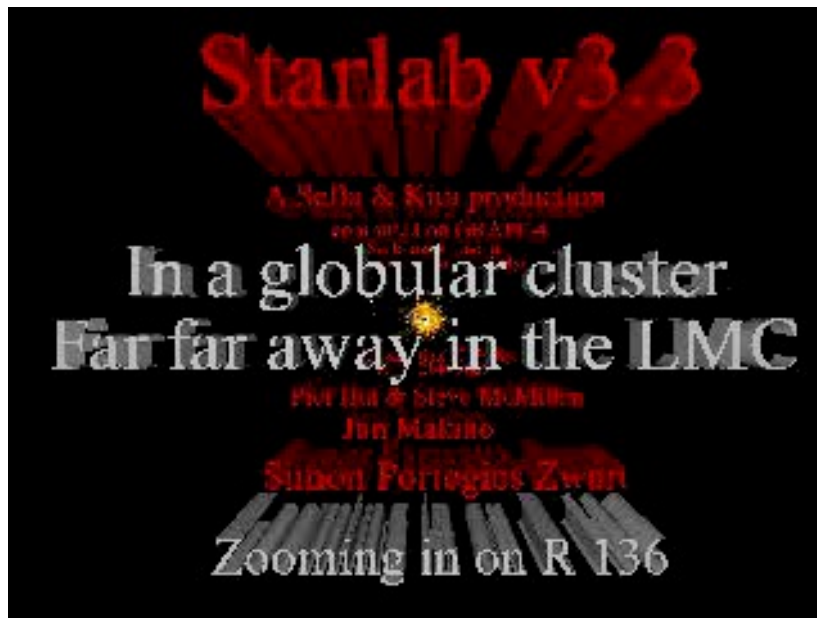
A special-purpose computer for gravitational many-body problems

Daiichiro Sugimoto*, Yoshihiro Chikada†, Junichiro Makino*, Tomoyoshi Ito*,
Toshikazu Ebisuzaki* & Masayuki Umemura‡

NATURE • VOL 345 • 3 MAY 1990

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In Univ. of Tokyo, computer science started as part of physics department (not math or CS)

Computer Physics Communications 60 (1990) 187–194

A special-purpose N -body machine GRAPE-1

Tomoyoshi Ito, Junichiro Makino, Toshikazu Ebisuzaki and Daiichiro Sugimoto
Department of Earth Science and Astronomy, College of Arts and Sciences, University of Tokyo, Tokyo 153, Japan

[CPC homepage](#)

Submit your first paper to *CPC*!

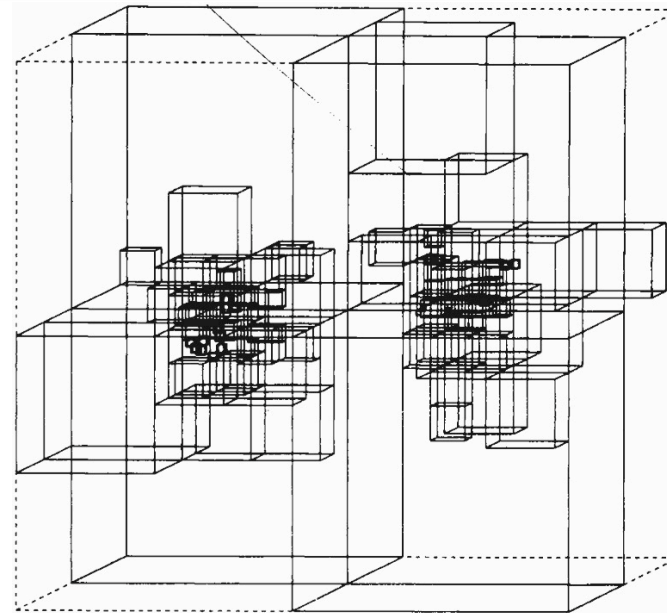
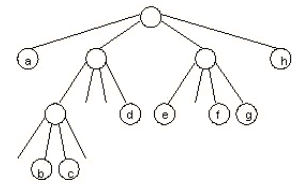
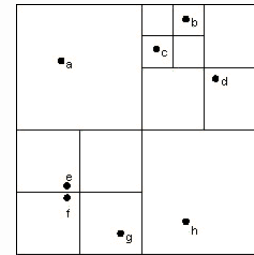
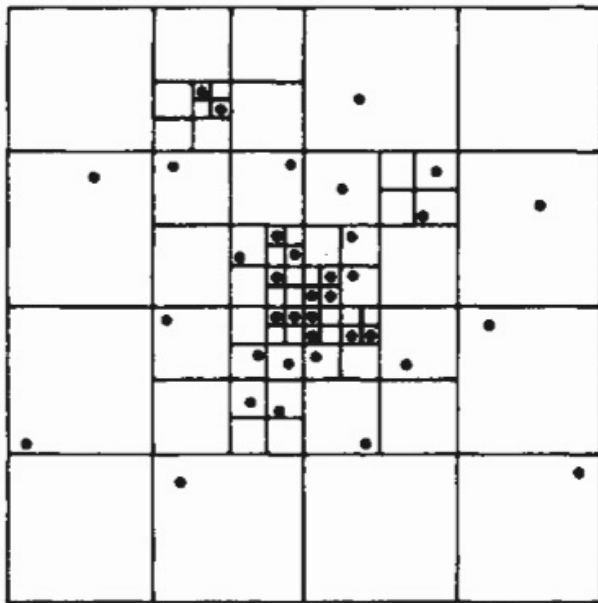
Enabling Science by Algorithm

NATURE

NATURE VOL. 324 4 DECEMBER 1986

A hierarchical $O(N \log N)$ force-calculation algorithm

Josh Barnes & Piet Hut



ACM Best Theses: Machine vs. Algorithm

DANNY HILLIS

Doctoral Dissertation Award
United States – 1985

CITATION

For his dissertation "*The Connection Machine*."

Watch: Hillis on Richard Feynman <https://www.youtube.com/watch?v=8CKW4A6jnJA>
<http://longnow.org/essays/richard-feynman-connection-machine/>



LESLIE GREENGARD

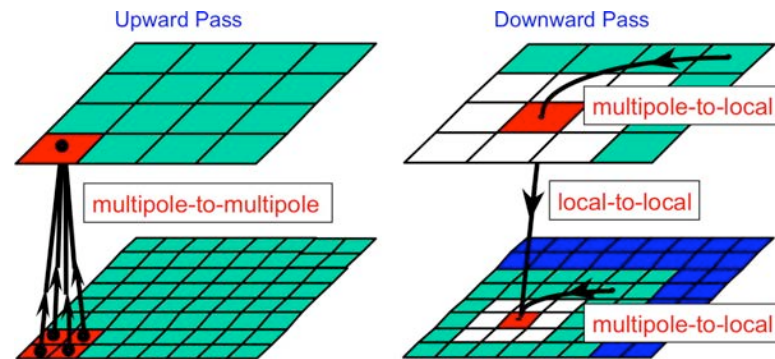
$O(N \log N) \rightarrow O(N)$

Doctoral Dissertation Award
United States – 1987

CITATION

For his dissertation "*The Rapid Evaluation of Potential Fields in Particle Systems*."

See lecture notes at <https://aiichironakano.github.io/cs653.html>



A Small Step in Parallel MD

VOLUME 71, NUMBER 1

PHYSICAL REVIEW LETTERS

5 JULY 1993

Structural Correlations in Porous Silica: Molecular Dynamics Simulation on a Parallel Computer

Aiichiro Nakano, Lingsong Bi, Rajiv K. Kalia, and Priya Vashishta

*Concurrent Computing Laboratory for Materials Simulations, Department of Physics and Astronomy,
Department of Computer Science, Louisiana State University, Baton Rouge, Louisiana 70803-4001*

(Received 9 November 1992; revised manuscript received 2 April 1993)

Molecular dynamics simulations of porous silica in the density range $2.2\text{--}0.1\text{ g/cm}^3$ are carried out on a 41 472 particle system using a multiple instruction multiple data computer. The internal surface area, pore surface-to-volume ratio, pore size distribution fractal dimension, correlation length, and mean particle size are determined as a function of the density. Structural transition between a condensed amorphous phase and a low-density porous phase is characterized by these quantities. Various dissimilar porous structures with different fractal dimensions are obtained by controlling the preparation schedule and temperature.

This work was supported by the U.S. Department of Energy, Office of Energy Research, Basic Energy Science, Materials Science Division, Grant No. DE-FG05-92ER45477. The computations were performed using the eight-node iPSC/860 in the Concurrent Computing Laboratory for Materials Simulations (CCLMS) at Louisiana State University. The facilities in the CCLMS were acquired with the Equipment Enhancement Grants awarded by the Louisiana Board of Regents through Louisiana



More N -body Simulations at SC

42 TFlops Hierarchical N -body Simulations on GPUs with Applications in both Astrophysics and Turbulence

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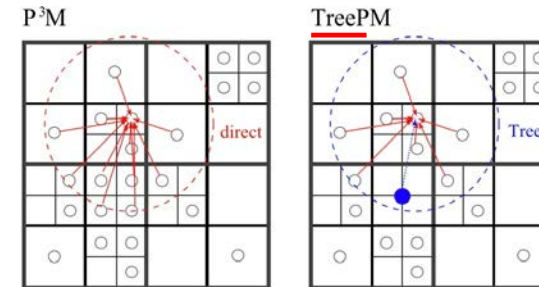
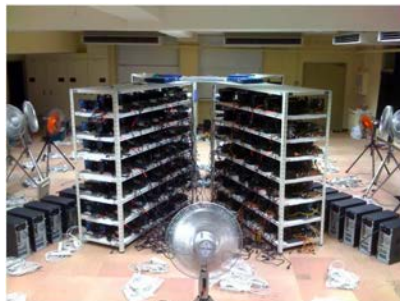
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Wako, Japan
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Makoto Taiji
High-Performance Molecular
Simulation Team
RIKEN Advanced Science
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taiji@riken.jp

2009 Gordon Bell Prize Price/Performance Category

Table 2: Price of the GPU cluster

Elements	Quantity	Price (JPY)	Price (\$)
GPUs	256	12,160,000	\$ 118,345
Host PCs	128	10,716,032	\$ 104,292
Network switch	4	644,800	\$ 6,275
Total		23,520,832	\$ 228,912



4.45 Pflops Astrophysical N -Body Simulation on K computer - The Gravitational Trillion-Body Problem

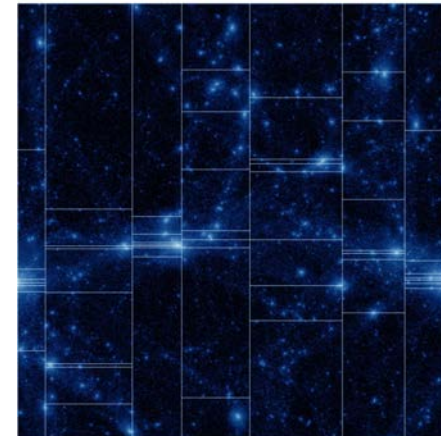
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IEEE/ACM supercomputing, SC12

*Machine
&
algorithm!*

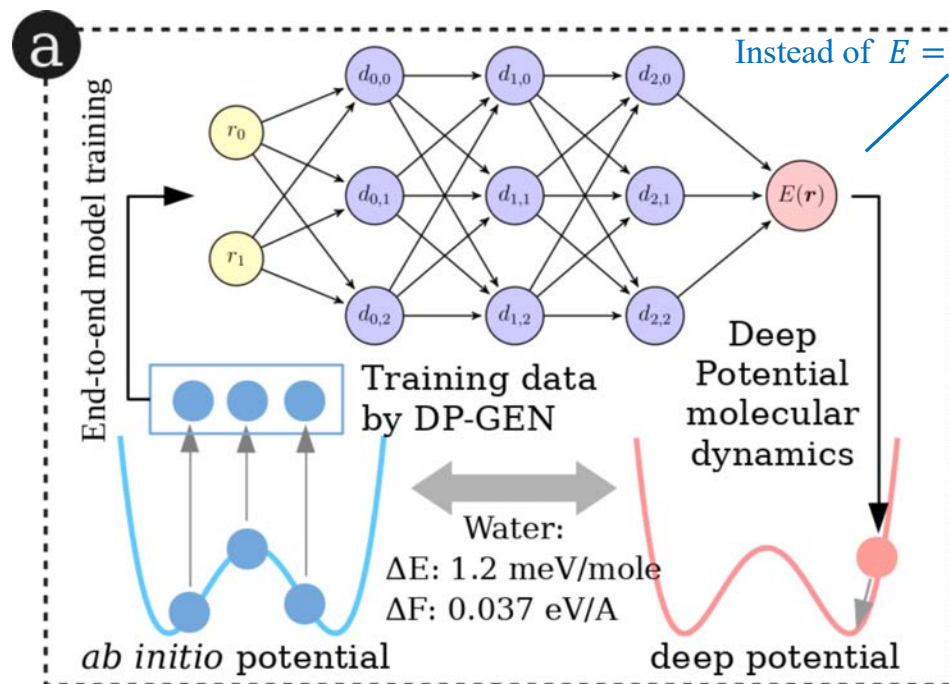


Now It's Deep MD

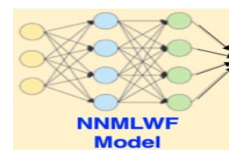
Pushing the limit of molecular dynamics with ab initio accuracy to 100 million atoms with machine learning

Weile Jia, Han Wang, Mohan Chen, Denghui Lu, Jiduan Liu, Lin Lin,
Roberto Car, Weinan E, Linfeng Zhang

Proc. Supercomputing, SC20, Gordon Bell prize

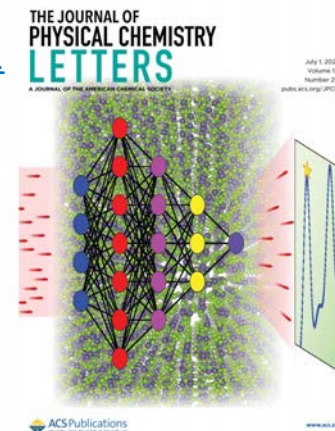


<https://aiichironakano.github.io/cs596/Jia-DeePMD-SC20.pdf>



Phys. Rev. Lett.
126, 216403 ('21)

J. Phys. Chem. Lett.
12, 6020 ('21)



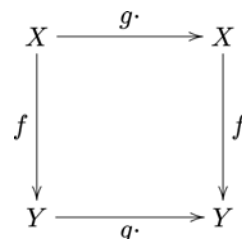
Accurate & Fast NNQMD: Allegro

- First principles-based neural-network quantum molecular dynamics (**NNQMD**) achieves quantum-mechanical accuracy with orders-of-magnitude less computation by replacing electronic wave functions by deep neural networks
- **Allegro (fast) NNQMD**: State-of-the-art *accuracy & speed* founded on *group-theoretical equivariance & local descriptors*

Musaelian et al., *Nat. Commun.* **14**, 579 ('23)

- An equivariant function f acts on an atomic geometry $x \in X$ under a group action g (e.g., rotation) as $f(g \cdot x) = g \cdot f(x)$

Thomas et al., *arXiv*. 1802.08219v3 ('18)



- Allegro speeds-up an equivariant message-passing (or graph neural-network) NNQMD, NequIP [Batzer et al., *Nat. Commun.* **13**, 2453 ('22)], by localizing atomic-environment descriptors, *i.e.*, restricting message passing to the nearest neighbor
- A complete set of atomic-environment descriptors is provided by atomic cluster expansion (ACE) [Drautz, *Phys. Rev. B* **99**, 014104 ('19)]; relationship between NequIP & ACE has been discussed in [Batatia et al., *arXiv*. 1802.08219v3 ('18)]

Power of Scale

- **Allegro-FM (foundation model)** for NNQMD described material properties & processes accurately, covering 89 elements in the periodic table, exhibiting *emergent capabilities* for which the model was not trained Nomura et al., *J. Phys. Chem. Lett.* **16**, 6637 ('25)
- **E3D: Allegro-FM trained *at scale*** learned interpretable chemical-bond *concepts* without supervision Hattori et al., *Nature Commun.* doi: 10.1038/s41467-026-74919-8 ('26)

E3D: Edge-wise emergent energy-decomposition framework ~ computational imaging to peek into AI brain

<https://viterbischool.usc.edu/news/2026/08/can-ai-understand-a-fundamental-concept-of-chemistry/>



P. W. Anderson, *Science* **177**, 393 ('72)
Nobel physics prize ('77)



<https://aiichironakano.github.io/cs596/Anderson-MoreIsDifferent-Science72.pdf>

Enabling Science by Online Game

nature

Vol 466 | 5 August 2010 | doi:10.1038/nature09304

LETTERS

Predicting protein structures with a multiplayer online game

Seth Cooper¹, Firas Khatib², Adrien Treuille^{1,3}, Janos Barbero¹, Jeehyung Lee³, Michael Beenen¹, Andrew Leaver-Fay^{2,†}, David Baker^{2,4}, Zoran Popović¹ & Foldit players



Ising Machine



1bit の世界の専用計算機 ——イジング・マシーン——

泰 地 真弘人
(東京大学教養学部)
(1994年 3 月 2 日受理)

Ising Machine:
A Special Purpose Computer for 1-bit Worlds

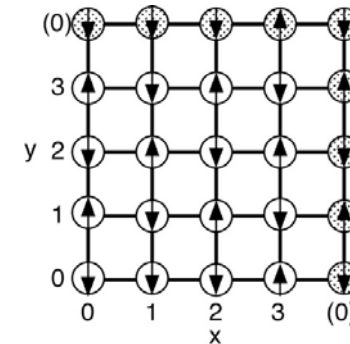
TAIJI Makoto
(Received 3 March 1994)

Abstract

This paper describes the development of special-purpose computer systems for Ising models, "Ising Machine" m-TIS 1 and 2. The first two sections explain Ising models and their Monte Carlo simulations. In section 3 and 4, I describe my motivation to build a special-purpose computer and the development of m-TIS 1. In section 5 and 6, the use of field-programmable gate arrays in a special-purpose computer is discussed. In the last two sections I discuss the potential abilities and future prospects of both Ising machine and a special-purpose computer in general. *J. Plasma Fusion Res.* **70**, 332 ('94)

cf. Original GRAPE was a 48-bit machine

$$V(s^N) = -J \sum_{(k,l)} s_k s_l - H \sum_k s_k, \quad s_k = \pm 1$$



<https://aiichironakano.github.io/phys516-lecture.html>

Inverse Ising model



USC Quantum Computation Center

- D-Wave 2X system with 1,098-quantum bits (qubits)

Phase transitions in a programmable quantum spin glass simulator

R. Harris^{1*}, Y. Sato¹, A. J. Berkley¹, M. Reis¹, F. Altomare¹, M. H. Amin^{1,2}, K. Boothby¹, P. Bunyk¹, C. Deng¹, C. Enderud¹, S. Huang¹, E. Hoskinson¹, M. W. Johnson¹, E. Ladizinsky¹, N. Ladizinsky¹, T. Lanting¹, R. Li¹, T. Medina¹, R. Molavi^{1,3}, R. Neufeld¹, T. Oh¹, I. Pavlov¹, I. Perminov¹, G. Poulin-Lamarre¹, C. Rich¹, A. Smirnov¹, L. Swenson¹, N. Tsai¹, M. Volkmann¹, J. Whittaker¹, J. Yao¹

Harris *et al.*, *Science* **361**, 162–165 (2018) 13 July 2018



Daniel Lidar



Bob Lucas

- Adiabatic quantum optimization

A background image showing a quantum circuit diagram with various gates and qubit lines. Overlaid on this is a yellow rectangular box containing the Hamiltonian equation:
$$H(t) = \sum_i \Delta(t) \sigma_x^i + \sum_i h_i \sigma_z^i + \sum_{i,j} J_{ij} \sigma_z^i \sigma_z^j$$
Below the equation, the terms H_I and H_F are indicated, corresponding to the first and last parts of the equation respectively.

http://www.isi.edu/research_groups/quantum_computing/home

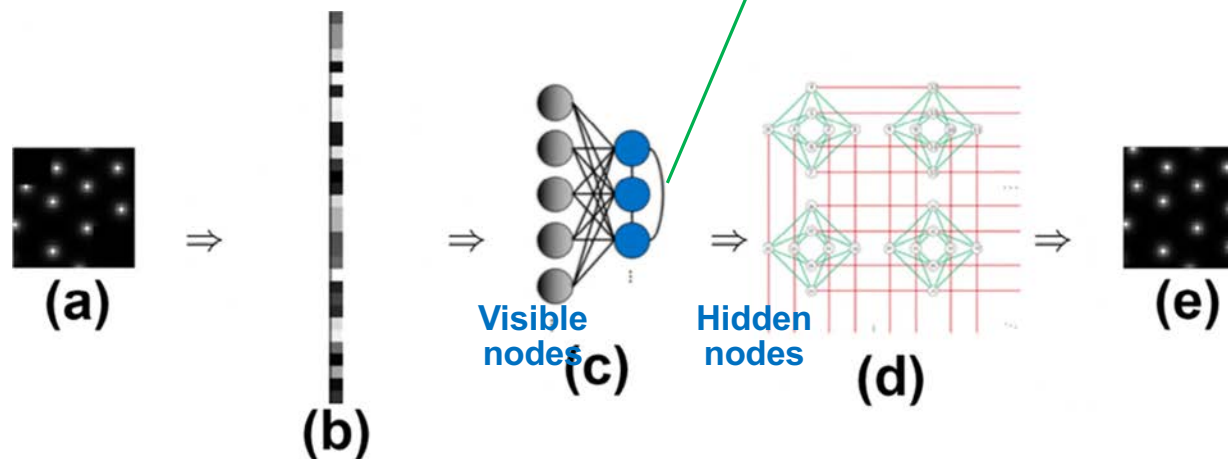
Machine Learning on D-Wave

Boltzmann machine modeling of layered MoS₂ synthesis on a quantum annealer

J. Liu, A. Mohan, R. K. Kalia, A. Nakano, K. Nomura, P. Vashishta, and K.T. Yao

[Comput. Mater. Sci. 173, 109429 \('20\)](#)

- Computing power of D-Wave allows *unrestricted* Boltzmann Machine to enhance machine learning performance



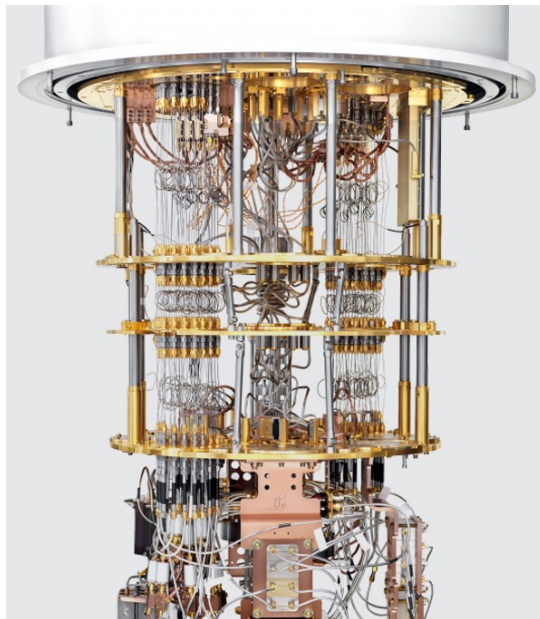
Restricted Boltzmann machine



Final project by Ankith Mohan (MSCS) with Jeremy Liu (PhD-CS)

Quantum Computational Science

- **Quantum computing for science: *Universal simulator of quantum many-body systems*** R. P. Feynman, *Int. J. Theo. Phys.* **21**, 467 ('82); S. Lloyd, *Science* **273**, 1073 ('96)]
- **Successfully simulated quantum many-body dynamics on publicly-available IBM's Q16 Melbourne & Rigetti's Aspen quantum computers** L. Bassman et al., *Phys. Rev. B* **101**, 184305 ('20), *ibid. B* **110**, 075116 ('24); *ibid. B* **113**, 235145 ('26)
- **AI-inspired domain-specific quantum compiler has reduced the circuit size by 30% below that by the vendor's native compiler** L. Bassman et al., *Quant. Sci. Tech.* **6**, 014007 ('21)



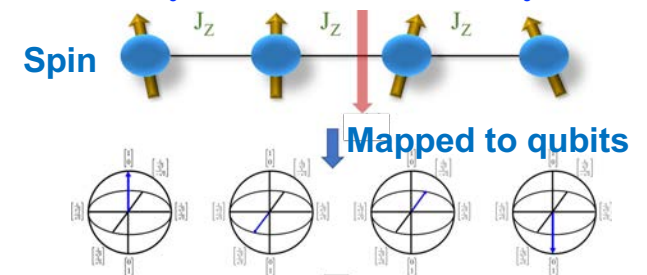
IBM Quantum Innovation Center at USC

<https://dornsife.usc.edu/qic/>

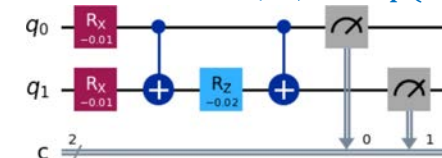
IBM agreement boosts USC's quantum computing leadership

The collaboration with IBM — part of USC President Carol Folt's Frontiers of Computing "moonshot" — establishes USC's IBM Quantum Innovation Center and propels USC researchers and students toward new discoveries.

March 11, 2024



Quantum circuit: $U(\Delta t) = \exp(-iH\Delta t)$



Quantum (Python) program

```
circ.rx(-2*dt*B, 0)
circ.rx(-2*dt*B, 1)
circ.cx(0, 1)
circ.rz(-2*dt*J, 1)
circ.cx(0, 1)
```